Hermes Lite 2 — Ethernet Recovery Firmware 7.4.3 Beta 1

Technical Review Paper

FOR TECHNICAL REVIEW ONLY — NOT INTENDED FOR CASUAL OPERATOR REVIEW

Experimental community firmware; not an official Hermes Lite 2 release

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1. Purpose and scope

This document is written for Steve Haynal, Chris Nichols, FPGA developers and other technically experienced reviewers. It is **not** the operator-facing README or a general installation guide. It records the design rationale, observed behavior, engineering limitations and build measurements for the first proposed public release of the independently developed 7.4.3 Beta 1 firmware.

The investigation began while preparing a Hermes Lite 2 for unattended remote operation. A physical Ethernet interruption could leave the radio inaccessible at its previous address, even though the HL2 remained powered. The practical objective is automatic recovery where possible, reducing occasions when an operator must travel to the remote station for a power reset.

The beta is offered as an optional experimental approach, not an official firmware replacement, and does not establish superiority over current upstream changes.

2. Reference implementation and modification scope

Pinned source reference: `softerhardware/Hermes-Lite2`, commit `3e467fd51572b4bbf9a28283c7562f1cb541e313`. On the HL2 target, its `hermeslite_core.v` advertises gateware version 74.2. This exact source revision was compiled independently for comparison; it has **not** been demonstrated byte-identical to a distributed official 7.4.2 RBF.

Changed RTL supplied for review:

• `gateware/rtl/ethernet/network.v` — network acquisition, timing, PHY reconnection and APIPA handling.

• `gateware/rtl/ethernet/dhcp.v` — DHCP exchange/reset behavior.

Only these two modified source files need to accompany the firmware binary for focused review. A full copy of the upstream source tree is unnecessary. A verified unified patch against the pinned upstream source has not yet been produced; reviewers should compare the supplied Verilog against that commit and classify all changed lines before considering adoption.

3. Intended behavior of Beta 1

1. A physical Ethernet reconnection triggers new network/DHCP acquisition and resets relevant DHCP transmit/receive state.

2. DHCP DISCOVER attempts occur at roughly three-second intervals over an approximately 30-second acquisition period.

3. If DHCP remains unavailable, the HL2 falls back to APIPA addressing (`169.254.x.x`).

4. APIPA operation remains stable without disruptive periodic DHCP retries after fallback.

5. Existing DHCP lease-renewal and fixed-IP functionality are intended to be preserved, but the static-IP path was not exhaustively verified in this beta.

Remote-operation limitation: if the HL2 has already entered APIPA but its physical Ethernet link remains up, restoration of the upstream DHCP service by itself does not trigger new DHCP acquisition. A qualifying physical link event or radio restart may still be needed. The firmware cannot restore a router, switch, Internet path or power supply that remains unavailable.

4. Relationship to newer upstream implementation

The subsequently reviewed upstream `master` revision includes a **five-second short-link hold** and reuses existing DHCP time registers, with a nominal **15-second DHCP acquisition timeout** and DISCOVER retransmissions around 1, 3 and 7 seconds after the initial request. Upstream commit `41d75b86e37f5a095971adc243d4a08e3ae9229d` is titled “Fix for lost IP when ethernet cable is removed.” A separate upstream commit, `f0247d4a85`, adds an October 25 testing release.

Beta 1 does **not** implement that five-second hold. It reinitializes on physical reconnection and offers a longer acquisition window. These are alternative design choices, with different effects on short link glitches and delayed DHCP service. Neither has been validated as preferable across all network conditions.

5. Hardware and network observations

Testing used one physical HL2, Thetis, Ethernet switches, DHCP service, Wireshark packet capture and ping. Reported observations include:

• DHCP DISCOVER retransmissions approximately **3.05 seconds** apart.

• APIPA fallback after approximately **30 seconds** without DHCP success.

• Continued APIPA response without repeated retry-induced network interruptions during observed tests.

• Successful acquisition after a physical link reconnection when DHCP was available, followed by resumed normal Thetis reception, without a radio power cycle.

• One unusual Ethernet reconnection delay of approximately **9–15 seconds**; cause remains undetermined.

Not established: comprehensive static-IP regression, all HL2 board/network variants, long-term unattended deployment, all DHCP renewal and power-restoration permutations, or a fault-free continuous timing analysis. Test observations should not be interpreted as blanket remote-reliability guarantees.

6. FPGA resource measurements

Both compilations used **Quartus Prime Lite 18.1**, target **Cyclone IV E EP4CE22E22C8**.

Resource	Upstream source, compiled Oct 26	Beta 1 development build, Oct 23	Delta (Beta 1 – upstream)
Logic elements	21,289 / 22,320	21,271 / 22,320	–18
Total registers	16,438	16,506	+68
Memory bits	424,464	424,464	0
9-bit multiplier elements	92	92	0
PLLs	2	2	0
Logic elements remaining	1,031	1,049	+18

Both whole-system designs consume approximately **95%** of available logic elements. The later build's reduction of 18 overall logic elements is **not proof that the Ethernet patch saves area**. Fitter placement, packing, synthesis decisions, configuration and differences elsewhere in the build can alter aggregate use. The October 23 report comes from the Beta 1 development process; its source/RBF has not been independently reproduced byte-for-byte from a clean checkout. The upstream baseline is a pinned source revision, not a verified byte-identical production binary.

An October 16 intermediate development report also exists (21,310 logic elements; 16,503 total registers), but must **not** be represented as stock production 7.4.2.

7. Timing and qualification

The original Beta 1 development documentation recorded a worst setup slack of about –**0.257 ns**. Negative setup slack signals an unmet reported requirement. The clean upstream source build completed fitting successfully but reported timing-constraint warnings, including incomplete setup/hold constraints. These are **not** complete timing sign-off results or evidence of equivalent margins.

Quartus build success and successful field tests do not establish timing closure. A before/after timing review with equivalent settings and a clean, reproducible build is necessary before declaring the change production-ready or recommending reliance on it for unattended remote operation.

8. Firmware identity and reproducibility

Firmware image: `HL2-DHCP-30s-COMPRESSED-2026-10-23.rbf`

SHA-256: `E83E70F14F6A94768411C3CE287CC02F54636A08B039E08F59C3A7C46153E641`

This checksum identifies the distributed test image. It does not prove that the accompanying two-file source snapshot is sufficient to recreate an identical RBF. The full upstream Quartus project, exact build settings, any generated outputs and a byte-for-byte validation would be required for reproducibility.

9. Technical review requested

Review should focus on:

• Whether `network.v` resets only what is necessary on a genuine link-reconnect event.

• Correctness of DHCP transmit/receive reset and transaction state after restart.

• DHCP scheduling and 30-second timeout implementation, including unexpected PHY changes mid-acquisition.

• APIPA continuity without background retries or receive/transmit disruption.

• Static-IP and normal DHCP lease-renewal regression.

• RTL state/resource footprint and opportunities to reuse counters without altering intended behavior.

• Full timing analysis, behavior across different routers/switches and prolonged unattended testing.

Credits: Original Hermes Lite 2 gateware by Steve Haynal, KF7O, and contributors. Chris Nichols identified important risks in earlier retry logic and contributed technical suggestions. Beta 1 development/testing was undertaken by Dalton, W5EIM, working collaboratively with ChatGPT.

Status: Technical-review document only. No merge request is implied, and this paper does not constitute approval for general or unattended deployment.